Practical 3.1 & 3.2: KoboToolbox Installation and Form Design

Self-hosted installation with kobo-install and Tanga flood survey form

Day 3 · 3 hours · Resilience Academy SDI Training

Reference: `crd.resilienceacademy.ac.tz` | Data: Tanga & Mwanza, Tanzania

Learning Objectives:

- Install a self-hosted KoboToolbox server using kobo-install
- Configure the server for local development
- Create an admin account and verify the installation
- Design a Tanga flood survey form with GPS coordinates

Prerequisites

- Docker and Docker Compose installed and working
- At least 4GB RAM available
- Ports 80 and 443 not in use (stop previous Docker services if needed: `docker compose down`)
- Python 3 installed

Step 1: Clone kobo-install

```
# Stop any running Docker services to free port 80
cd ~/my-crd && docker compose down

# Clone kobo-install
cd ~
git clone https://github.com/kobotoolbox/kobo-install.git
cd kobo-install

# Check Python version
python3 --version # must be 3.x
```

Step 2: Run the Setup Wizard

```
python3 run.py
```

Answer the wizard prompts as follows:

```
Installation type? --> Local (press 1 + Enter)
Local domain name? --> localhost
Use HTTPS? --> No
Advanced options? --> No
```

The wizard generates a `.kobo/envfiles/` directory with all configuration files. You can inspect them later.

Step 3: Start KoboToolbox

```
# Start all services (downloads images first time, ~10 min)
python3 run.py --start

# Monitor startup --- open a second terminal and run:
cd ~/kobo-install
docker compose logs -f kpi
# Wait for: Listening at http://0.0.0.0:8000

# In another terminal, watch kobocat
docker compose logs -f kobocat
# Wait for: Listening at http://0.0.0.0:8001

# Check all containers
docker compose ps
# Should show: kobocat, kpi, postgres, mongo, redis, nginx all Up
```

Step 4: Create Admin Account

```
# Create superuser
docker compose exec kpi python manage.py createsuperuser
# Enter: username=admin email=admin@example.com password=Admin1234!

# Access KoboToolbox:
# Main interface: http://localhost
# Django admin: http://localhost/admin
```

Step 5: Design the Tanga Flood Survey Form

In KoboToolbox web interface:

1. Click **New** → **Build from scratch**
2. Name your form: `Tanga Flood Survey 2024`
3. Add these questions:

```
Question: GPS Location | Type: Point (geopoint)
Question: Flood depth (m) | Type: Decimal number
Question: Date observed | Type: Date
Question: Ward name | Type: Text
Question: Photo of site | Type: Photo
Question: Affected households | Type: Integer
Question: Water source | Type: Select one
  Options: River, Rain, Pipe, Other
Question: Infrastructure damage | Type: Select multiple
  Options: Road, Bridge, Building, Electricity, None
Question: Observer notes | Type: Text area (multiline)
```

```
# Deploy the form
# Click: Settings > Sharing
# Set: Anyone can view this form (for testing)
# Click: Deploy

# Submit test data
# Click Preview (web form)
# Enter Tanga GPS: latitude=-5.0700, longitude=39.1000
```

```
# Fill other fields with test data
# Submit 3 records
```

Checkpoint:

- ✓ `docker compose ps` shows all KoboToolbox containers as "Up"
- ✓ `http://localhost` loads the KoboToolbox login page
- ✓ Admin account login works
- ✓ Tanga flood survey form deployed successfully
- ✓ 3 test submissions visible in Data → Table view
- ✓ GPS coordinates visible in Data → Map view